

Analog Electronics – Unit 1: Operational Amplifier

Exam Notes

These notes cover the core Unit 1 topics required for typical B.Tech Analog Electronics exams. Focus on circuit diagrams, formulas, derivations, and key definitions.

1. Ideal Operational Amplifier Characteristics

- Infinite open-loop voltage gain ($A \rightarrow \infty$).
- Infinite input impedance (no current enters input terminals).
- Zero output impedance (can drive load without voltage drop).
- Infinite bandwidth.
- Infinite Common Mode Rejection Ratio (CMRR).
- Zero input offset voltage.
- Zero noise and zero drift.

2. Inverting Amplifier

Input signal is applied to the inverting terminal through resistor R_1 . Feedback resistor R_f connects output to the inverting input. Non-inverting terminal is grounded.

Virtual Ground Concept: Due to negative feedback, $V_+ \approx V_-$. Since V_+ is grounded, $V_- \approx 0$.

Derivation:

- Input current: $I = V_{in} / R_1$
- Same current flows through R_f because input current into op-amp ≈ 0
- $I = -V_{out} / R_f$
- $V_{in} / R_1 = -V_{out} / R_f$
- **Voltage Gain:** $A_v = V_{out} / V_{in} = -R_f / R_1$

3. Non-Inverting Amplifier

- Input is applied to non-inverting terminal.
- Feedback network connected to inverting terminal.
- Voltage at inverting terminal equals input voltage due to virtual short.

Voltage Gain: $A_v = 1 + (R_f / R_1)$

4. Op-Amp Integrator

- Feedback component is a capacitor instead of resistor.
- Output voltage proportional to integral of input signal.

Output Equation: $V_{out} = -(1/RC) \int V_{in} dt$

5. OpAmp Differentiator

- Input component is capacitor and feedback is resistor.
- Output proportional to rate of change of input.

Output Equation: $V_{out} = -RC (dV_{in}/dt)$

6. Comparator and Schmitt Trigger

- Comparator compares two input voltages.
- If $V_{in} > V_{ref} \rightarrow$ output saturates to positive supply.
- If $V_{in} < V_{ref} \rightarrow$ output saturates to negative supply.
- Schmitt trigger is a comparator with positive feedback.
- It introduces hysteresis (two switching thresholds) to reduce noise.

7. Important OpAmp Parameters

Parameter	Definition
CMRR	Ratio of differential gain to common-mode gain. Higher value means better no
Slew Rate	Maximum rate of change of output voltage (V/ μ s).
Input Offset Voltage	Small differential input voltage required to make output zero.
Gain Bandwidth Product	Product of amplifier gain and bandwidth; constant for opamp.

Exam Tip: Practice derivation of inverting amplifier gain and remember formulas for integrator, differentiator, and noninverting amplifier. These commonly appear in exams.